

General Relativity

Pre-final Examination Spring 2022 Total Possible Score: 100

Name: _____ Class: _____ Student Number: _____

1. (25 points) Consider the following line element:

$$ds^2 = \frac{a}{z^2} (dx^2 + dy^2 + dz^2), \quad z > 0, \quad a = \text{constant}.$$

- (a) Calculate all Christoffel symbols.
- (b) Calculate all the non-zero terms of the Riemann curvature tensor.
- (c) Calculate the terms of the Ricci tensor.
- (d) Calculate the Ricci scalar.
- (e) Is this a maximally-symmetric space? Explain why or why not.
Explicitly test this for one term using the relevant equation for curvature given in the textbook.

2. (20 points) Show by explicit calculations whether or not each one of the following vectors, that is, rank-(1,0) tensors, is a Killing vector for the 2-sphere S^2 with line element

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$$

- (a) $S = (\cos\phi, -\tan\theta \cos\phi)$
- (b) $W = (\cos\phi, -\cot\theta \sin\phi)$

3. (55 points) Let's consider a two-dimensional space, a sub-manifold of $\mathbb{R}^3$ formed by the surface of a paraboloid:

$$x = \rho \cos\phi \quad y = \rho \sin\phi \quad z = a\rho^2 \quad \rho = \sqrt{x^2 + y^2} \quad a = \text{constant}$$

(So $d\rho$ measures movement straight up the 'bowl', and $d\phi$ measures movement around a horizontal circle of constant ρ .)

- (a) Find the metric/line element in terms of the coordinates ρ and ϕ .
- (b) Find all Christoffel symbols.
- (c) Find the geodesic equations. Write as differential equations parameterized by a scalar λ .
- (d) Find all non-zero elements of the rank-(1,3) Riemann curvature tensor, and all non-zero elements of the rank-(0,4) Riemann tensor.
- (e) Calculate all elements of the Ricci tensor and the Ricci scalar.
- (f) Write one of the geodesic equations as a total derivative. Integrate this introducing the integration constant $\mathcal{C}$.
- (g) Insert the result from (f) into another geodesic equation to obtain a equation with only a single variable.
- (h) Find the parameterized velocity and acceleration of a particle traveling straight up the side of the paraboloid ($d\phi = 0$).
- (i) Using results from the preceding three parts, show that this particle is traveling along a geodesic.
- (j) (a little difficult) Suppose a vector, initially parallel to the curve $\rho = \text{constant}$ ($d\rho = 0$), is parallel-transported once around the $\rho = \text{constant}$ curve on the paraboloid, from $\phi = 0$ to $\phi = 2\pi$.

What is the angle between the vector's new direction after it goes around once and the vector's initial direction?

